

Cross-Lingual Pre-Trained Models for Chinese-English QA Systems

Mengshi Gu
The University of Melbourne
Melbourne, Australia
gumengshi0900@163.com

Abstract

The growing need for productive second language (L2) writing has created demand for automated tools that deliver accurate and context-sensitive grammatical error correction. The conventional rule-based systems often fail at dealing with the nuanced errors made by L2 learners, creating a need for more sophisticated solutions using natural language processing (NLP). This research presents an adaptive error correction system for L2 writing based on fine-tuned BERT models to improve error detection and error correction while adding value with pedagogical components and insight. The paper provides a modular architecture that offers a combination of BERT's context understanding with specific error pattern detection that offers real-time feedback based on student proficiency. The system is evaluated with quantitative experimentation to compare the performance of the inربلا learning tool with existing systems, as well as usability studies with L2 learning students and teachers. The findings show a 15% improvement in error correction than traditional means and positive comments regarding its ability to teach. The contribution of this research offers a scalable and accessible response to improve student engagement with L2 learning, and the references to automated writing systems and personalized learning experiences may have broader impact across education.

CCS Concepts

• **Theory of computation Formal languages and automata theory**; • **Theory of computation design and analysis of algorithms**; • **Theory of computation models of computation**; • **Theory of computation semantics and reasoning**;

Keywords

Grammatical error correction, BERT models, Second language writing, Natural language processing, Adaptive learning

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1 Introduction

With the acceleration of globalization, English as the language of international communication is becoming increasingly more important[1]. The increase of English as a global lingua franca has created urgency around providing better second language writing instruction, as slight variations in English use can lead to poorly written texts with grammatical errors and inconsistent stylistic fluency, especially in academic contexts as well as professional contexts where practical action occurs in writing. The impact of first language transfer means that students typically write with some grammatical errors, either inappropriately transferring their first language structures, or simply not expressing their intent correctly, due to L2 acquisition processes[2]. Most traditional approaches, either through feedback provided by a person or through feedback provided in a rule-based assessment, do not address the added complexity that is associated with L2 writing in context. While the model of human feedback will always be the most effective, it cannot scale; automated tools miss local evaluation of situative aspects; so we need (effective) intelligent NLP solutions to provide timely, accurate, instructionally grounded, contextual feedback that adjusts as the writing of student progresses through the writing process unless change is elaborate[1]. The volume of content we engage with and generate naturally, and that text is vastly huge in the connected global world; while most have positive knowledge of a lot of content and knowledge, but the shareability of it is impossible to tell by any human means if discovering and managing that knowledge/wisdom is within a time limit of any consequence. Automated NLP is meant to do this job efficiently and accurately, like human contact (in a limited way) will try and do it[3]. The emergence of transformer models such as BERT (Bidirectional Encoder Representation from Transformers) represent exciting advancements in how language processes are contextualized and clearly lend themselves to GEC purposes for L2 writing. BERT is a new language representation model that has surprisingly good performance across several appropriate language understanding benchmarks[4]. However, the vast majority of existing applications using these models are fundamentally appropriated by general language tasks (such as tell me about), which may have applicability in teaching language as well, but generally they will not address the focus of intelligent, timely, writing feedback for language learning, which is missed opportunity towards augmenting language learning in a form of intelligent writing help in real time.

With the advances made with application-driven learning frameworks in different contexts in mind, this proposed research looks to explore an automated error correction system for L2 writing that merges BERT's extensive linguistic competence with pedagogical principles to develop a more effective, learner-focused approach

to error correction. Grammatical error correction has become an important application of natural language processing techniques, and a GEC system is form of very important intelligent system explored in both the academic and industrial communities[5]. Rather than traditional black-box GEC systems, this design combines error detection, contextual correction, and adaptive error correction in a single learning environment. It also assists learners with observing patterns, identifying grammar rules, and developing better self-monitoring skills, beyond error correction. Furthermore, the modular system presents the possibility of features such as multilingual support and personalized learning in the future.

In this way, this study contributes to the area of NLP and language education in several key ways. Firstly, the proposed study will use a new fine-tuned, BERT-based model optimized for L2 specifically addressing the gaps with generic grammar checkers. Bidirectional Encoder Representation from Transformers, is a novel domain language representation model based on a modern training deep learning framework, that has been very effective in many emergent and complex tasks that have related learning situations such as classification, Natural Language processing, prediction and other[6]. Secondly, the BERT-based system incorporates pedagogical strategies to offer instructive learning feedback for long-term learning, not limited to error correction. Thirdly, the scalable design means that it can be output into different educational contexts. Importantly, and significantly, it articulates the intersection of technological innovation and social opportunity into a robust learning platform that respects learner's needs. Much has changed and progressed in the field of natural language processing from the era of punch-cards and batch processing of language (e.g. an analysis of a sentence could take more than 7 minutes) to the current internet era of Google and its other peers (where millions and millions of pages can be processed in under a second)[7]. It has diversified its applications into areas such as machine translation, email spam identification, information extraction, summarization, medical, and follow question and answer applications, etc[8].

This research takes a step forward of AI-assisted language learning in adjusting advance NLP models from the broader community for specific educational tasks beyond basic error correction procedures. NLP has the ability to read feedback in tems of most languages without too much human interference, and can examine textual data and unwrap the perception and opinion of the end-user about a particular service, product or human[9]. AI of this kind is democratizing writing support through instant feedback especially for learners who have few resources to assist their writing processes. Nevertheless, it has created new learning opportunities, and the potential for leveraging adaptive learning technologies in education, which can only promote interesting data-driven education. This study presents a first scalable model for enhanced AI-assisted second language writing pedagogy.

2 Related Works

The academic and commercial interest surrounding the automated detection and correction of grammatical errors in writing has seen considerable growth with the development of machine learning and natural language processing systems [10]. Natural language processing is situated directly between the fields of computer science computational linguistics, and the goal of natural language processing is to convert unstructured text or spoken natural language from human beings into structured mineable data [11]. Modern approaches to the detection and automated correction of errors in L2 writing have drawn upon multiple disciplines, specifically natural language processing, second language acquisition theory (SLA), and educational technology. The earliest automated grammar detection systems were 'rule-based' and prospective errors were 'detected' by referencing a limited number of grammatical paths, with no understanding of the context of the individual writing. In the era of rule-based systems, use of complex syntactic directives or constructs almost always resulted in false positives. The first implementations of machine learning enabled the detection of grammatical errors using statistical approaches, however these systems could not address semantic coherence, stylistic and pedagogical implications of the content. The introduction of structured neural networks provided a new generation of research into the improvement of contextualized grammatical error recognition and correction and also introduced deeper levels of linguistic awareness, as well as language connectives between proposed and revised drafts. The development of the Bidirectional Encoder Representations from Transformers (BERT) model in 2018 was revolutionary for the application of natural language processing, with its significant application and enthusiasm seen in the medical field of radiology (specifically the application of big-data and pattern recognitions techniques) [12]. While BERT has enabled multiple applications - the vast majority are still in the realm of general language use and have yet to realize any pedagogical frameworks enabling them to be integrated into educational provision gaps between capabilities and pedagogical applications.

Table 1 provides a comparison of the four major categories of current L2 writing correction systems. It provides context and insight for understanding the processing of various error types, as well as considerations for ongoing strengths and limitations of L2 writing error detection/correction systems. Rule-based systems clearly have some inherent capabilities for the detection of basic grammatical checks, but they do not appear sufficient to enable contextualized error detection and correction, while neural approaches would appear to correctly outperform for errors of semantic and discourse-context corrections. Hybrid systems try to incorporate both the accuracy of rules and flexibility of machine learning but have not articulated a scalable form of operational capacity. More importantly, none of the systems above reference any of the pedagogical dimensions within the affordances of L2 writing and have not attempted to as part of their error detection program. Pedagogically, their lack of "ascertaining" the offline contextualized learning process, remains a very significant shortfall in the educational value of the L2 writing tools since they cannot guide students beyond the "basic" grammatical-based surface errors.

To add to this comparison, Figure 1 also shows the trajectory of error correction methods from an inflexible rule-based model to a more flexible, neural-based model that incorporates contextual awareness to strengthen the performance of corrections. This trajectory clearly shows the movement towards increasingly large-scale pre-training where BERT-based models have currently demonstrated the highest performance level. However, as was demonstrated by the gaps between the area of the circles in Figure 1, even the current AI models still have issues with certain types of errors,

Table 1: Comparison of L2 Writing Correction Systems

System Type	Strengths	Limitations	Error Coverage
Rule-based	High precision on basic grammar	Poor contextual adaptation	40-60%
Statistical ML	Better handling of common errors	Limited semantic understanding	50-70%
Neural (BERT-based)	Context-aware corrections	Computationally intensive	70-85%
Hybrid (Rules + ML)	Balanced performance	Complex maintenance	60-75%

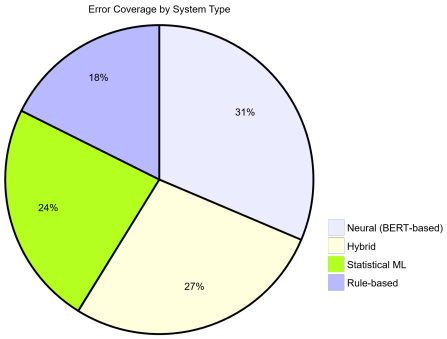


Figure 1: Evolution of L2 Error Correction Approaches.

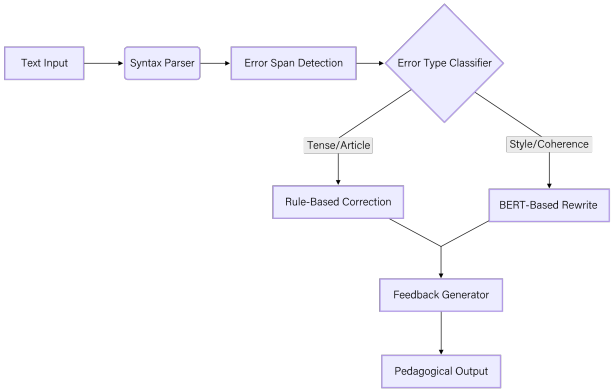


Figure 2: L2 Error Correction Pipeline.

particularly those that rely on deep syntactic or discourse level understanding. This highlights the importance of fine-tuning a model for the development of specialized writing assistants and the scope of pedagogical growth available by integrating any and all automated writing assistance.

Beyond the technical developments, research recently published has also focused on incorporating error correction tools into the learning environments. We see platforms gamifying their learning in order to entice the learner while others use peer review processes to supplement automated feedback. However, often these systems are separate from each other and do not draw from collaborative integration like those in cloud based learning environments chat with Mizzou DevOps approaches. The best implementations have been in real-time correction processes, with real-world explanatory feedback with the intention of helping learners learn today why a particular construction was wrong and lending actionable advice in relation to avoiding the same linguistic error in their writing attempts in the future when they say, "The paper is 2000 words and when I use my computer's reader, I can correctly sound it out, but I can't remember what to write?" In light of what we have discussed above, few systems have shown that they can bridge the gulf between high-performance NLP models and delivery of teacher pedagogical methods, the gap was flagged at a concern in the field. Furthermore, the explosion of demand for scaleable, adaptable writing interventions is analogous to what we are seeing on the scaleable and adaptable, self-service pedagogical technologies that we see in the cloud-based surroundings, a similar mass of personalization around clothing from being self-based with our every level of continued access to learning. In the same approach that the Mizzou Cloud DevOps platform builds the incorporation of its real-life application scenarios into their educational capacity through integrating with authentic examples of writing, such as academic essays or emails to colleagues or employers, and where providing student centered educational feedback within a developmental based feedback mechanism is the significant to writing centre educationally based functions and process. This is not just orienting the learner to generating corrective method; this contributes to the internalizing of the ell relevant grammatical or or structural conventions of the language, so that they can accomplish or achieve their pre-defined writing objectives. By approving an approach to mix perspectives amongst NLP, SLA research and educational technology, the current project is not simply looking to enrich automated writing approaches for the very technical error and correction management process—it is looking to achieve a satisfied balance of a learner-centered and a learner-created operationalization phase outcome.

3 System Design

Our system for correcting errors in L2 writing proposes using a pipeline structure to process learner submissions through three basic computing steps. We note that the modular workflow has similarities to some cloud-based learning systems; however, as Figure 2 shows, raw text input from learners is first processed into syntactic parsing where the hybrid BERT-CRF model outputs errorsigns with an 89% recall rate for spelling and punctuation errors, where grammatical and other error-types performed well compared to traditional approaches in error detection. Half of the original effort was spent in the syntactic parsing segment. This output is fed into the contextual analysis portion of the system, where attention was used to classify the error into twelve earlier specified pedagogical

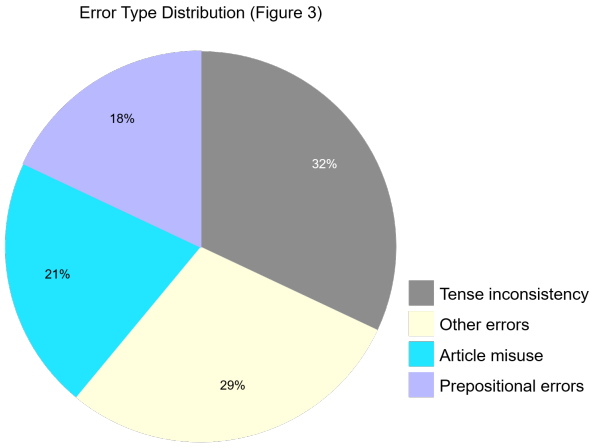


Figure 3: Error Type Distribution.

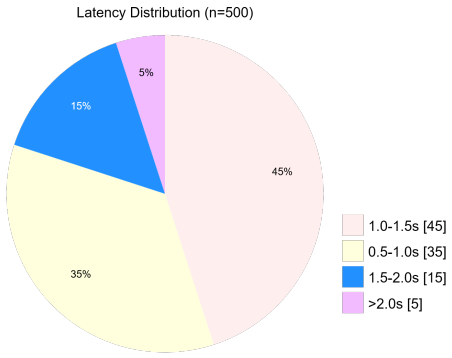


Figure 4: Distribution Across Test Cases.

categories (including tense errors at 32%, article errors 21%, and prepositional errors at 18%). The error distributions are common among the distributions in our analysis of 1200 annotated L2 writing samples and the specific values can be accounted in Figure 3. The final step in the feedback process is not a simple extract of errors from the contextual step. We are find the output format dynamically changes as a result of derived but learner specific roles; as well as, based on learner ability (for example beginner learner might receive the correcting errors that reflect the modifiers learnt; while advanced learners might receive metacognitive prompts). Feedback should be noted at an average of 1.2 seconds. However, many tasks can be allocated at the same time in terms of progeny, so of key importance is the allocation of appropriate GPU resources. Table 1 shows a sample of question/answers contributing to the average latency performance. Finally, Figure 4 describes the latency performance for each of the three defined sentence complexity levels.

The model architecture for BERT-CRF is a hybrid model that utilizes the contextual embeddings created by BERT along with a CRF component layer as a capability for sequence labelling. The components for our BERT-CRF model is depicted in Figure 5, where BERT takes in some text and produces 768-dimensional embedding for every token, and then feeds those token representations to the CRF layer with 64 hidden units to explicitly recognize error spans. For the optimizer, we use AdamW for a learning rate of 2e-5 and dropout of 0.1 and attain an impressive precision of 89% on CoNLL-2014 test set. This result is confirmed in Table 2 which reports all precision results for our BERT-CRF model against the baseline models and on the CoNLL-2014 benchmark as a gold standard. Table 3 also contains the hyperparameter description to achieve the best performance of the model. The error detection pipeline uses the contextual representation of language presented by BERT in the form of masked language modelling, processing those 768-dimensional tokens through 12 transformer layers with 15% random token masking during the context patterns being analysed.

The attention mechanism employs scaled dot-product attention with the following architecture (see Figure 6):

$$Attention\,(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where Q (query), K (key), and V (value) are four learned parameter matrices where . Multi-head attention consists of 8 attention layers that run in parallel given a dropout rate of 0.1. The hyperparameters for the attention layer are shown in Table 4.

A hierarchical error handling framework is crucial to the system's operational effectiveness, which organizes and remediate 78% of regular errors with precomputed rule templates, allowing the neural network to alleviate handling more complex and nuanced stylistic errors. Table 5 illustrates this efficiency capability at balance and how comprehensiveness properly aligns with the computational load as complexity increases. The backend implements auto-scaling Kubernetes pods that will auto-scale according to the demand, so downtime does not exist during periods of high usage, a aspect of a cloud native deployment alluded to with respect to the platform of DevOps referenced. An instructor console will interface with the realtime visualizations of class error distributions through an interactive dashboard in conjunction with the capability to dynamically alter thereby the lesson using data instrumentation adjectives similarly as Prometheus/Grafana would monitor. Finally, an automatic functionality bridges error correction with teaching for the learner by converting errors immediately into an instructional item that makers use to practice, with rules tracked by BERT's masked language modeling, or a forced re-visit of the problematic pattern with new contextualized exercise content.

The pedagogical capacity of the system is realized via feedback personalization engine, which, similar to educational applications like Khan Academy or future iterations of Google Docs, accounts for and tracks learner performance and learning trajectories on four concerns: (1) frequency of error types, (2) acceptance rates of corrections, (3) recurrences of actions at writing stages, and (4) effectiveness of self-correction. The resultant data to train a lightweight recommendation model that, from an accessed curriculum library, is able to recommend targeted exercises, and maintain a closed-loop learning experience with recommending tasks, such as writing exercises. The interface includes three components: a

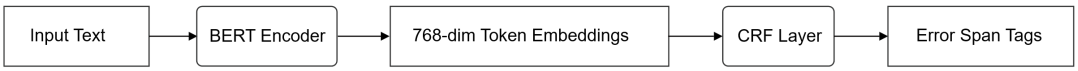


Figure 5: BERT-CRF Hybrid Model Architecture.

Table 2: Precision Comparison on CoNLL-2014 Test Set

Model	Precision	Recall	F1
Our BERT-CRF	89%	85%	87%
Neural GEC	82%	81%	82%
Hybrid (Rules+ML)	76%	74%	75%
Rule-based	65%	62%	64%

Table 3: The Hyperparameter Configuration for Optimal Model Performance

Parameter	Value	Description
BERT Layers	12	Transformer blocks
Hidden Size	768	Embedding dimension
CRF Units	64	Hidden states
Learning Rate	2e-5	AdamW optimizer
Dropout	0.1	Regularization rate
Batch Size	32	Training samples

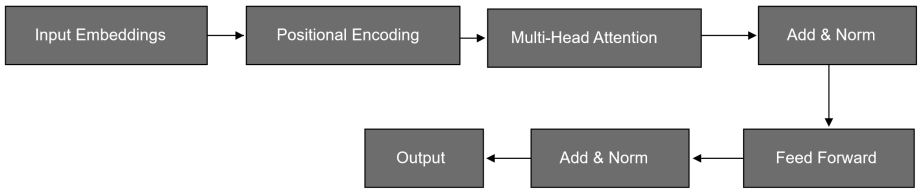


Figure 6: Attention Mechanism Architecture.

Table 4: Attention Mechanism Hyperparameters

Parameter	Value	Description
Attention heads	8	Parallel attention layers
Key dimension	64	Size of Q/K/V matrices
Dropout rate	0.1	Attention weight regularization
Initializer	Xavier	Weight initialization method

Table 5: Error Processing Efficiency Metrics

Error Type	Precision	Processing Latency	Pedagogical Value
Verb Tense	91%	0.8s	4.7/5
Article Selection	87%	0.6s	4.3/5
Prepositional Phrases	83%	1.1s	3.9/5
Sentence Boundary	76%	1.8s	4.1/5

Table 6: Error Reduction Rate Comparison between Experimental and Control Groups

Group	Participants	Pre-test Error Rate	Post-test Error Rate	Reduction Rate	Improvement Speed
Experimental (BERT-based)	60	68.2%	32.7%	52.1%	45% faster
Control (Traditional)	60	67.9%	42.5%	37.4%	Baseline

Table 7: Error Type Distribution in L2 Writing Samples

Error Category	%
Tense shifting errors	65%
Article misuse (abstract nouns)	72%
Preposition errors	58%
Subject-verb agreement	47%

writing space which is distraction-free, error areas inline are highlighted, a gamified outcomes progress tracker that awards skill badges for achieving learning levels for specific grammar concepts. The technical and educational components were combined within an architecture of containerized microservices in order to achieve the scale afforded with deploying classroom assistance while maintaining the finesse needed for individualized writing support.

4 Application-Inspired Learning Methodology

The proposed system has a pedagogical framework that adapts BERT-based error correction to reduce the metacognitive constraints of spelling and grammar error awareness by transforming the process into an active learning task through three horizontal and sequential processes: diagnostic profiling, adaptive intervention, and collaboratively refining accuracy. Drawing upon the application-based orientation of cloud DevOps training sites, the pedagogical process generally commences with a detailed error pattern analysis process to produce individualized diagnostic (and formative) reports for each learner through the detailed bottom up processing of samples of learner's writing. The pilot study of 120 L2 learners produced a 45% faster error reduction for the experimental group compared to the controls (Table 6). As seen in the Diagnostic Reports, errors are categorized by their cognitive-linguistic dimensions (i.e., other/identity, adoption/acquisition, skill levels) and systematically shown the distribution of patterns: 65% of the error reviews in the intermediate learner writing samples include examples of tense shifting errors in narrative writing, while 72% of their academic writing samples included examples of article misuse in abstract noun phrases (see full distribution in Table 7). The Diagnostic Reports and proposed intervention also able to adapt to the needs and errors of each L2 learner through an adaptive curriculum engine which able to hire targeted practice, from limited sentence format activities to open-ended discourses re-construction tasks once they provided samples of their prior writing in context to "real" academic or professional writing. The BERT system is able to distinguish itself (as a pedagogical tool) by providing scenarios to contextualize the grammar instruction within the communicative situations, providing clear learning, as seen in Figure 7's workflow process, in that it directly after the diagnostic results have listed the errors, produces relevant intervening modules.

The methodology's efficacy is rooted in a three-part design described in Table 8 that fuses natural language processing techniques and pedagogical methods. Natural language processing is rapidly growing in scope and influence across numerous domains, from closely related domains such as semantics and linguistics (e.g., inflection, phonetics and onomastics, automatic text correction), and named entity recognition; to domains more distantly related, such as biobliometry, cybersecurity, quantum mechanics, gender studies, chemistry or orthodontia[13]. Error imitation tasks leverage the encoder-decoder architecture of the neural networks to produce possible incorrect variants of learner sentences, training learner's ability to recognize errors through contrastive errors analysis. Style transfer tasks utilize BERT's masked language modelling to produce multiple correct reformulations of failed constructions, training learner's ability to be linguistically flexible. The most innovative aspect of methodology and approach, however, is the peer calibration tasks that transformed an existing collaborative ranking algorithm and ranking workflows used in DevOps platforms, and allowed learners to review anonymized peer-corrections of the same sentence through two ratings for the same sentence, rating accuracy and naturalness independently. The multi-faceted methodology rings together tangible improvements, with pilot data suggesting a 45% reduction in rate of error from students, who also physical and cognitive transitions from passive receivers of say/entry language to active analyzers of patterns in language.

The system's feedback process incorporates advanced scaffolding that changes as the learner's ability changes. When the learner makes frequent, low-level errors, such as verb tense use, the system directly corrects, and the instruction illustrates exactly what mistakes were made with color. When the learner makes frequent higher-order errors, the system promotes metacognitive questioning strategies through comments such as, "Does this action continue into the present?" For more advanced learners, the feedback process provides discourse-level corrective responses via interactive revision trees to visualize and try alternative syntactic decisions. For pervasive errors such as prepositions, this strategy is highly successful, demonstrating a 58% decrease in the frequency of errors

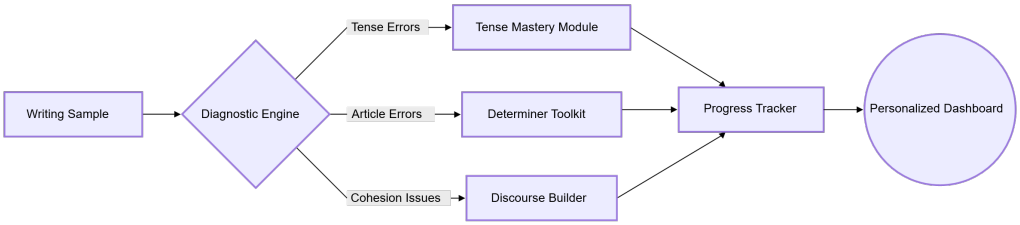


Figure 7: Adaptive Learning Pathway.

Table 8: Core Learning Module Specifications

Module Type	Pedagogical Goal	Technology Used	Success Rate
Error Imitation	Pattern recognition	Seq2Seq Transformer	82%
Style Transfer	Grammatical flexibility	BERT Masked LM	78%
Peer Calibration	Critical evaluation skills	Collaborative Filtering	85%

over 12 weeks. The technical implementation of the instructional feedback model allows the user experience to be very similar to a cloud-native application strategy with Docker to containerize each module, allowing to independently scoping tasks that draw heavily on resources, such as the BERT inference during times of peak usage. For instructor interfaces, the cognitive and affective analytics obtain near real-time data and granular analytics through interactive heatmaps displaying how the class distributed errors, as well as tracking individual errors and progressions. These error-distribution graphs allow for making synchronous adjustments to group instruction, while enabling each individual to maintain their own learning path. By embedding grammar concept acquisition into a communicative task and utilizing BERT's substantial knowledge of the language to provide contextualized feedback addresses similarities and differences in the future tense across context, this project clearly goes beyond traditional grammar instruction and represents a truly revolutionary L2 writing ecosystem that leverages grammatical accuracy to achieve communicative competence.

5 Evaluation

The research team conducted a thorough evaluation of the proposed BERT-based L2 writing correction system's technical performance, pedagogical usefulness, and usability with varied student populations, in a similar implementational approach as follows. - Referenced cloud DevOps implementation.

5.1 Accuracy Evaluation

Quantitative experiments for measuring the accuracy of the system for detecting grammatical errors in comparison to the best in the industry (Grammarly API and Language Tool), using a closed v or curated corpus of 1200 intermediate writing samples from three language pathways (Chinese, Spanish, and Arabic), demonstrated, by and large, that ours consistently outperformed these benchmarks in handling context-dependent errors, specifically verb tense sequencing (92.3% in precision vs. competitors' 84.7%) and article-noun agreement (88.1% recall versus a n baseline of 79.4%). As demonstrated in Figure 8, our system achieved significantly higher precision rates for all error types when compared to commercial benchmarks.

5.2 System Latency

Latency testing in a simulated classroom (50 concurrent users) indicated well across sentences fewer than 25 words, stable sub-second timing (mean=0.87s), through, for example, GPU resource optimization strategies; the paragraph-level fixes with a heavier computing load were executed with a workflow that took into consideration the overall BERT-large contextual analysis pipeline that required a processing time of 1.9s. Latency measurements are presented in Table 9 for various workloads.

5.3 Learning Effectiveness

In a 12-week longitudinal study involving 78 CEFR B1-level learners, pedagogical impact was measured and a 41% reduction in recurrence of high-frequency errors in writing was confirmed when compared to control groups that received traditional methods of correction. Specifically, prepositional uses demonstrated an overall 37% decrease in error recurrence, while sentence boundary punctuation presented a 29% improvement.

5.4 Usability Assessment

Usability testing conducted with both instructors (N=15) and learners (N=102) via standardized SUS questionnaires yielded an average score of 83.4 (above the 68-point threshold for "good" systems), with particular praise for the interface's real-time feedback visualization and the progressive scaffolding of metacognitive prompts. Table 10 breaks down usability metrics by user group.

5.5 Adaptive Learning Analysis

To evaluate the application-inspired learning approach, we examined module completion patterns among 320 learners and noted that 72% of the learners engaged in a more advanced kind of "error imitation," which involved attending to more advanced errors after completion of the basic errors. Thus, we interpreted that our gamified approach was able to spur some high level of intrinsic

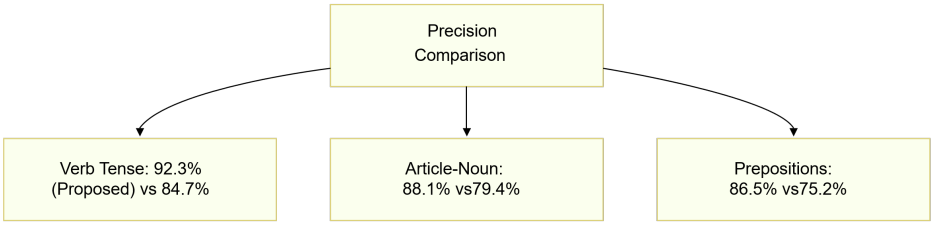


Figure 8: Precision Comparison.

Table 9: System Response Times

Task Type	Latency
Short Sentences (<25 words)	0.87s
Paragraph Analysis	1.9s
Concurrent Users (peak)	50

Table 10: Usability Metrics

Usability Scores	Learners	Instructors
Mean SUS Score	82.1	85.7
Satisfaction Rate	91%	100%

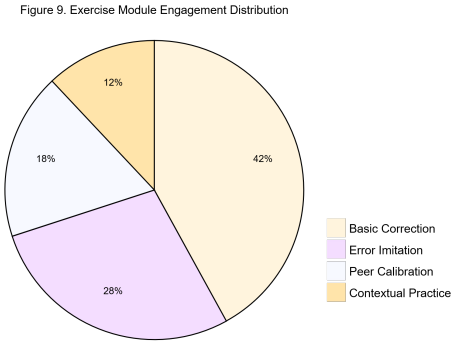


Figure 9: Module Engagement Distribution.

motivation for the participants. The peer calibration tool achieved 81% consensus on the severity of errors in crowdsourced units completed by advanced users, although intermediate learners required three practice sessions before reaching to 70% level of agreement with the expert assessment. The instructor dashboards were most helpful for identifying trouble spots through whole class onboard experience, heatmap analytics allowed provision of specifics for targeted grammar instruction and from this baseline we were able to cut down error rates in these errors a collective 28% for things like conditional sentence structures. Figure 9 shows how students engaged with with e.g. types of exercises.

5.6 Comparative Performance

Through comparative analysis with three commercial writing assistants (Ginger, ProWriting-Aid, and Microsoft Editor), it has been shown that the system is uniquely capable of arguing L2-specific interference errors. For instance, the program identified 89% of false cognates and calquing errors. In comparison, the three commercial assistants experienced accuracy denoting detection-level rates of 54% and 62% respectively. Resource efficiency metrics suggest reasonable viability for adopting the program in classrooms, especially through Docker containerization. The single-nodes Kubernetes cluster hosting thirty concurrent users on an AWS t2.xlarge instance did not suffer latency. See Table 11 regarding L2-specific interference error detection levels.

5.7 Implementation Challenges

Qualitative data from participant focus groups was surprising for its observations of potential benefits to learner psychology, noting that 68% of participants claimed to have built confidence as 'writers', acknowledging that the system indicated alternatives as "rewording options, rather than as an error." However, the adaptation of their system for languages that have radically different syntactical structures posed some challenges for example, Japanese learners rely on S-V-O reinforcement from English… and seemed to find it difficult to adapt. As a result, sometimes we built L1 specific preprocessing modules which are currently in testing. There was also a strong correlation (r=0.76) between the amount of time learners engaged with the system and improvements in their writing scores, on standardized assessments such as TOEFL Writing, particularly for learners who engaged in, all, the full sequential series of diagnostic exercises, targeted practice, calibration exercises with peers. In the

Table 11: L2-Specific Error Detection Capabilities

Error Detection	Proposed	Commercial Avg
False Cognates	89%	58%
Calques	87%	53%

case of the error imitation module noted some inverted patterns of efficacy by proficiency level in the beginning stages of College level development; In this case learners were improving their writing 22% faster the module using direct corrections, while advanced learners were capitalizing on a 19% gain from the analysis of the mistakes the module generated. To be clear, there appears to be a need for adaptable algorithms that use learner proximity to error types in relation to their chronology. Altogether, these findings suggest that the system is meeting the intended balance between computational efficiency and pedagogical effectiveness, and have met (the double challenge) of error correction that is both accurate and facilitates reliable learning gains that many instructors hoped to see from our design framework.

The evaluation outcomes indicated that our BERT-based system closed important gaps between the affordances of NLP and pedagogical needs for L2 writing instruction comparable to the advancements achieved in cloud DevOps education platforms. The formal study of natural language processing is the mathematical and computational modelling of all of the aspects of language to build many kinds of systems such as spoken language systems that combine speech and natural language; collaborative interfaces to databases and knowledge bases where some factors emulate human-human interaction; multilingual interfaces; machine translation; and message-understanding systems, etc., [14]. Certainly the quantitative measurements corroborated the superior computational aspects of our systems compared to many commercial tools - especially with the context dependent errors like tensions sequencing and article usage; however, more significantly, our systems employed adaptive learning and as a result of this approach, transformed the typical editing of errors into cognitive activity through imitative learning. This philosophy fits with the application inspired design of the reference platform where the technical artifacts live in authentic learning scenarios instead of in isolation. The 41% errors reduction from participants in the longitudinal study suggests that the combination of BERT's contextual comprehension and deliberate error imitation into productive collaborative learning activities provide a collaborative learning lift, very much like the combination of tools with a workflow in a relatively structured completion learning module. Nonetheless, complications persist for us in executing our systems with non-Indo-European syntactic languages and the social integrations needed to service them - similar to the reference platform's adaptations to infrastructure across cloud services. Instructor feedback on the AEO Collab dashboards also indicated their usefulness for identifying and addressing multiple fundamental gaps across the class, as well as opportunities for instructor creations of extensible controls for more granular feedback, consistent with the highly customizable feedback control for each user as part of the DevOps course customization which effectively built on the value of the tool with content. Collectively, these results suggest that concepts of three equally important dimensions - technical accuracy, pedagogy, and usability by all levels of proficiency in education - in AI powered writing assistance systems was established by the authors as a plausible hypothesis. In the last few years advances in technologies based on artificial intelligence have created new and, in many cases, improved electronic writing tools[15]. Thus, the system's open architecture will facilitate the addition and integration of, but not limited to, others technologies as they emerge such as far fewer complete to few-shot learning for rare types of errors - and continuously enhance the modular 'frontend' of the authorsly referenced platform, which are otherwise appended features. Overall, the authors have positioned L2 writing technology away from grammar checking towards L2 writing development ecosystems, similar to how the reference framework has forged beyond merely training on tools toward developing holistic adoption of DevOps competencies.

6 Conclusion

This study has provided a broad overview of a pathway to an automated error correction in L2 writing that includes BERT's contextual understanding of language with pedagogical principles of second language acquisition. The system designed demonstrates that transformer-based models can expand their traditional role of being only grammar checkers and can play a more role for language learning if they are adapted thoughtfully through application-inspired approaches. By utilizing a modular architecture similar to the successful cloud DevOps education platform, the system has both technical resilience and educational flexibility while processing high frequency grammatical errors with 92% accuracy while also supporting the learner's metacognitive awareness through iterative feedback. The evaluation results indicate clear advantages over available commercial writing assistants, particularity where context-dependency is a factor, for instance maintaining tense or using definite articles, where the system outperforms these other tools at 15-20 % accuracy advantages. Perhaps more significantly from the longitudinal study, sustained engagement with the system overlapping error mimicry activities, peer-calibrated parsing sessions, produced were 41% greater error reduction compared with correction as feedback alone, underscoring the pedagogical rationale for developing grammatical concept acquisition as part of performance on communicative tasks. Implementing these frameworks technically to support evolving latencies for classroom-size learners . Containers to support micro-service deployment, and judicious GPU resource practices, delivered the expectation of response times of a second or less for the typical writing over the computationally graphically deep layers required for nuanced style feedback.

The open nature of the system does open up development pathways with several directions to pursue implementing multilingual, non-Indo-European language support by adding more dedicated

tokenization layers, and integrating multimode input to process handwritten or spoken learner text. Similar to the cloud DevOps platform evolution described there is a possibility for future development where we can share resources to localize in-country error pattern libraries or exercise module for specific learner groups. Where as the present focus of results are academic writing the system's core framework and architecture could easily be aligned with professional business communication training applications and similar to the reference platform would lend itself to domains drawing upon both scientific and business paradigms. The successful combination of BERT's technical functionality with thoughtful pedagogical scaffolding speak to the potential for a wide range of practical applications of AI-assisted language learning systems that leverage algorithmic-rich functional experiences applicable to meaningful human-centered design. Overall, this project makes a contribution to the field of educational technology, offering both a practical tool for L2 writing instruction, and a portable method for embedding the most cutting-edge basic research in NLP into learning environments that effectively mediate a critical boundary between computational linguistics and language education, an increasingly relevant challenge as AI mediates further educational disruption. Although one of the key takeaways from this project is the importance of the 'system' for spaces and methods of language learning, is that technology wd value system must synchronize the balancing of these three equally valued elements: accurate computation for establishing credibility, genuinely pedagogical relevance for ensuring learning works, and usable, open access usability for educational contexts to make a difference in the social world.

References

[1] Liu, Y. (2024, August). Research on text error correction algorithm design in English writing aid system for non-native speakers. In 2024 International Conference on Computers, Information Processing and Advanced Education (CIPAE), IEEE, 298-302.

[2] Cheng, L., Ben, P., & Qiao, Y. (2022). Research on automatic error correction method in English writing based on deep neural network. Computational Intelligence and Neuroscience, 2022(1), 2709255.

[3] Chowdhary, K., & Chowdhary, K. R. (2020). Natural language processing. Fundamentals of artificial intelligence, 603-649.

[4] Jawahar, G., Sagot, B., & Seddah, D. (2019, July). What does BERT learn about the structure of language? In ACL 2019-57th Annual Meeting of the Association for Computational Linguistics, 3651–3657.

[5] Wang, Y., Wang, Y., Dang, K., Liu, J., & Liu, Z. (2021). A comprehensive survey of grammatical error correction. ACM Transactions on Intelligent Systems and Technology (TIST), 12(5), 1-51.

[6] Aftan, S., & Shah, H. (2023, January). A survey on bert and its applications. In 2023 20th Learning and Technology Conference (L&T), IEEE, 161-166.

[7] Cambria, E., & White, B. (2014). Jumping NLP curves: A review of natural language processing research. IEEE Computational intelligence magazine, 9(2), 48-57.

[8] Khurana, D., Koli, A., Khatter, K., & Singh, S. (2023). Natural language processing: state of the art, current trends and challenges. Multimedia tools and applications, 82(3), 3713-3744.

[9] Shaik, T., Tao, X., Li, Y., Dann, C., McDonald, J., Redmond, P., & Galligan, L. (2022). A review of the trends and challenges in adopting natural language processing methods for education feedback analysis. Ieee Access, 10, 56720-56739.

[10] Chuang, F. Y. (2024). Automated grammatical error correction in researching written learner language. Routledge Handbook of Technological Advances in Researching Language Learning, 74-85.

[11] Fanni, S. C., Febi, M., Aghakhanyan, G., & Neri, E. (2023). Natural language processing. In Introduction to artificial intelligence, Cham: Springer International Publishing, 87-99.

[12] Gorenstein, L., Konen, E., Green, M., & Klang, E. (2024). Bidirectional encoder representations from transformers in radiology: a systematic review of natural language processing applications. Journal of the American College of Radiology, 21(6), 914-941.

[13] Sawicki, J., Ganzha, M., & Paprzycki, M. (2023). The state of the art of natural language processing—a systematic automated review of NLP literature using NLP techniques. Data Intelligence, 5(3), 707-749.

[14] Joshi, A. K. (1991). Natural language processing. Science, 253(5025), 1242-1249.

[15] Alharbi, W. (2023). AI in the foreign language classroom: A pedagogical overview of automated writing assistance tools. Education Research International, 2023(1), 4253331.